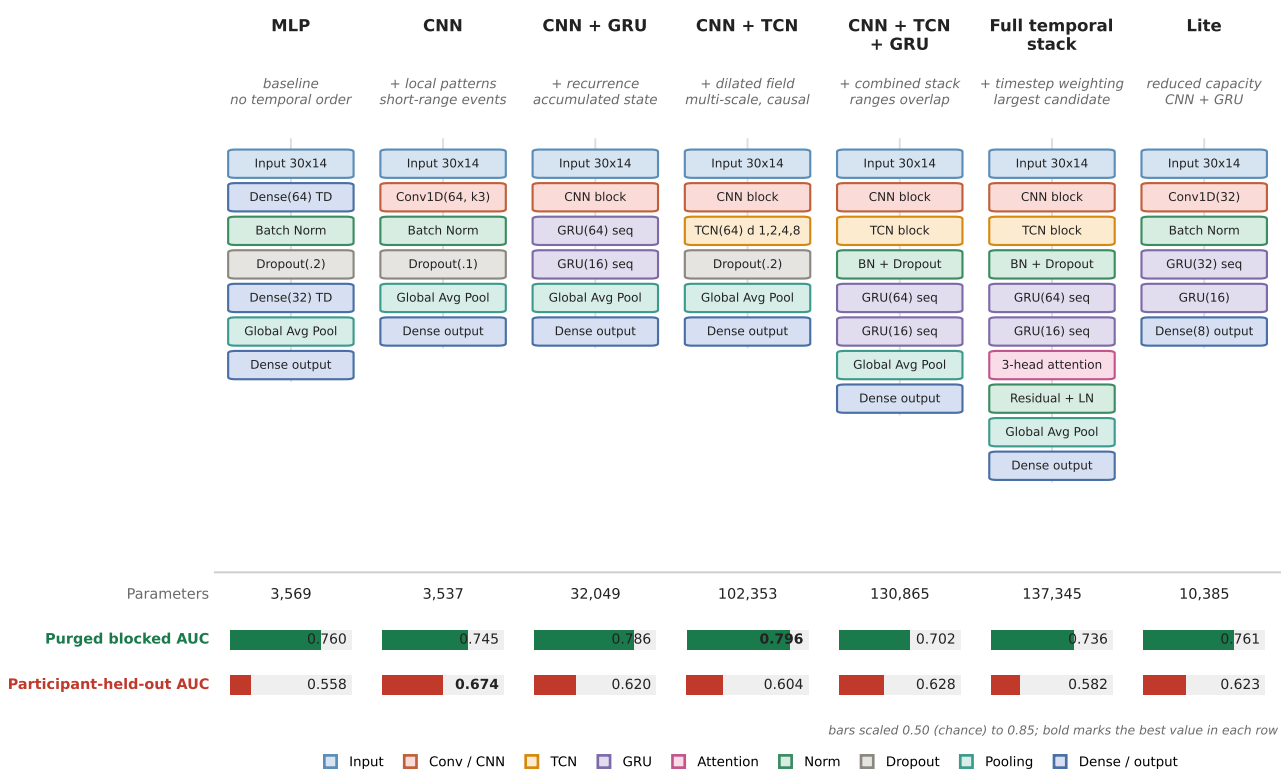
MLP
baseline
no temporal order
Input 30x14
Dense(64) TD
Batch Norm
Dropout(.2)
Dense(32) TD
Global Avg Pool
Dense output
CNN
+ local patterns
short-range events
Input 30x14
Conv1D(64, k3)
Batch Norm
Dropout(.1)
Global Avg Pool
Dense output
CNN + GRU
+ recurrence
accumulated state
Input 30x14
CNN block
GRU(64) seq
GRU(16) seq
Global Avg Pool
Dense output
CNN + TCN
+ dilated field
multi-scale, causal
Input 30x14
CNN block
TCN(64) d 1,2,4,8
Dropout(.2)
Global Avg Pool
Dense output
CNN + TCN + GRU
+ combined stack
ranges overlap
Input 30x14
CNN block
TCN block
BN + Dropout
GRU(64) seq
GRU(16) seq
Global Avg Pool
Dense output
Full temporal stack
+ timestep weighting
largest candidate
Input 30x14
CNN block
TCN block
BN + Dropout
GRU(64) seq
GRU(16) seq
3-head attention
Residual + LN
Global Avg Pool
Dense output
Lite
reduced capacity
CNN + GRU
Input 30x14
Conv1D(32)
Batch Norm
GRU(32) seq
GRU(16)
Dense(8) output
Parameters
3,569
3,537
32,049
102,353
130,865
137,345
10,385
Purged blocked AUC
0.760
0.745
0.786
0.796
0.702
0.736
0.761
Participant-held-out AUC
0.558
0.674
0.620
0.604
0.628
0.582
0.623
bars scaled 0.50 (chance) to 0.85; bold marks the best value in each row
Input
Conv / CNN
TCN
GRU
Attention
Norm
Dropout
Pooling
Dense / output